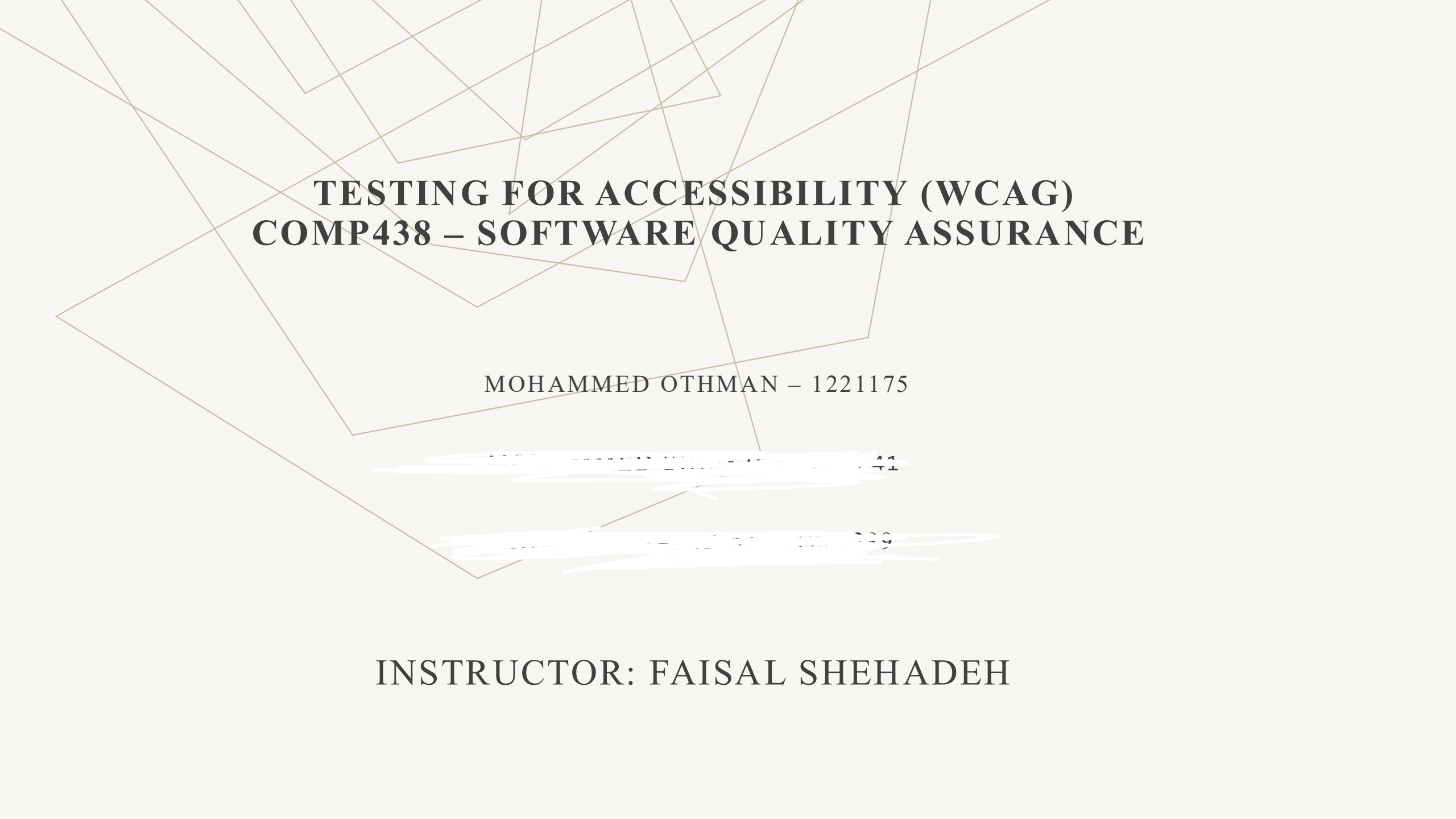


The background of the slide features a series of thin, light-brown lines that intersect to form various geometric shapes, including triangles and polygons, creating a complex, abstract pattern.

TESTING FOR ACCESSIBILITY (WCAG)

COMP438 – SOFTWARE QUALITY ASSURANCE

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ACCESSIBILITY IN PROGRAMMING

- ❑ Accessibility means making websites and applications easy to use for all people
- ❑ Helps people with disabilities and other users
- ❑ Benefits users with mobile devices, slow internet, or small screens
- ❑ Improves understanding and usability of digital content

WHO NEEDS ACCESSIBILITY AND WHY

- ❑ People with visual, hearing, physical, or learning disabilities
- ❑ Older adults with age-related difficulties
- ❑ Users with temporary or situational challenges
- ❑ Accessibility ensures equal access and fairness for all users



WCAG OVERVIEW

- ❑ WCAG: Web Content Accessibility Guidelines
- ❑ International accessibility standards
- ❑ Developed by the World Wide Web Consortium (W3C)
- ❑ Used as the main reference for accessibility testing



METHODS & TOOLS

ACCESSIBILITY TESTING METHODS

- ❑ Manual testing using keyboard and screen readers
- ❑ Automated testing using accessibility tools
- ❑ Hybrid approach combining manual and automated testing
- ❑ Each method has strengths and limitations

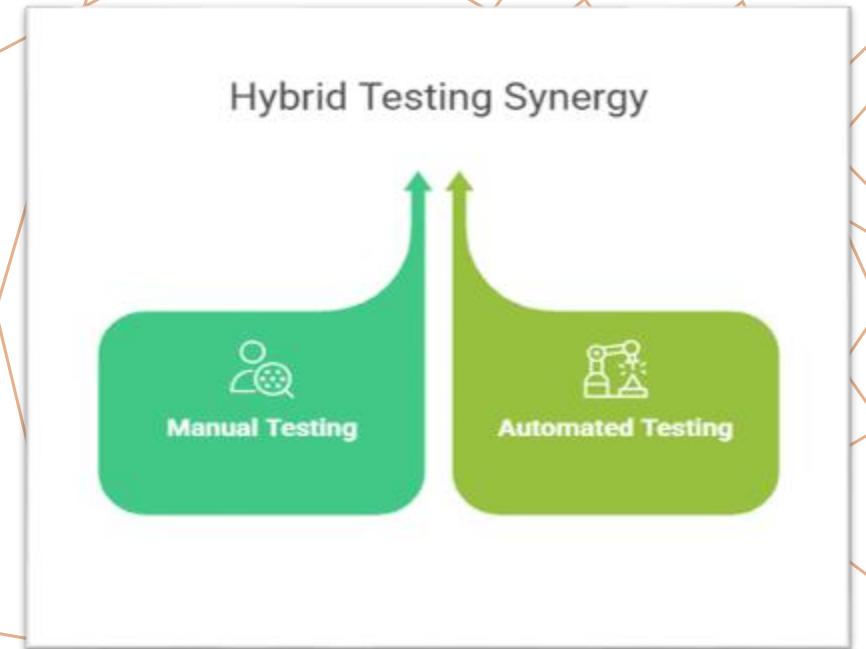


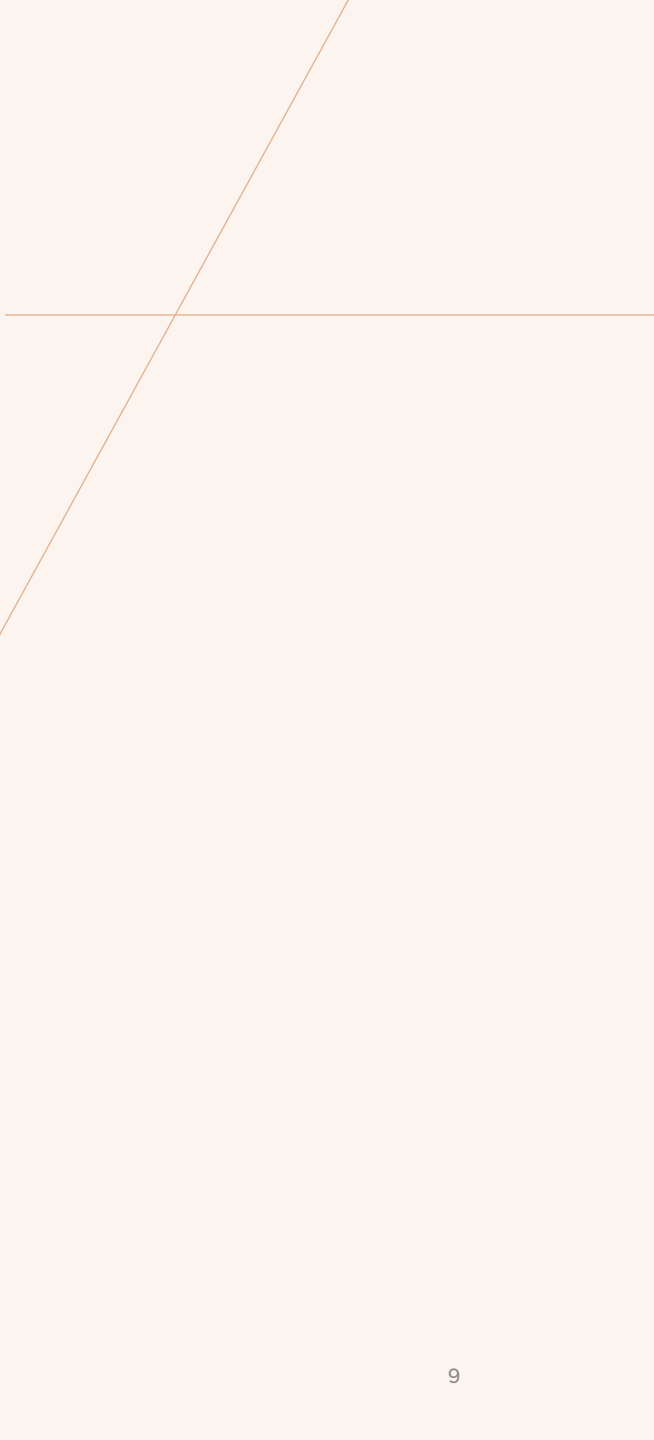
ACCESSIBILITY TESTING TOOLS

- ❑ WAVE: highlights accessibility issues on web pages
- ❑ Axe: detects WCAG violations and supports automation
- ❑ Screen Readers (NVDA, VoiceOver): test real user experience
- ❑ Lighthouse: provides accessibility score and quick checks

IMPORTANCE OF HYBRID TESTING

- ❑ Automated tools are fast but limited
- ❑ Manual testing detects usability and context issues
- ❑ Hybrid testing provides better accessibility coverage
- ❑ Best practice for WCAG compliance





ANALYSIS, IMPACT AND FUTURE

CHALLENGES IN ACCESSIBILITY TESTING

- ❑ Many accessibility checks require human testing
- ❑ Large and dynamic websites are hard to test
- ❑ Third-party content may change frequently
- ❑ Applying WCAG to non-web systems is challenging

**Challenges with Accessibility
Guidelines Conformance and Testing,
and Approaches for Mitigating Them**

W3C Working Draft 19 June 2020

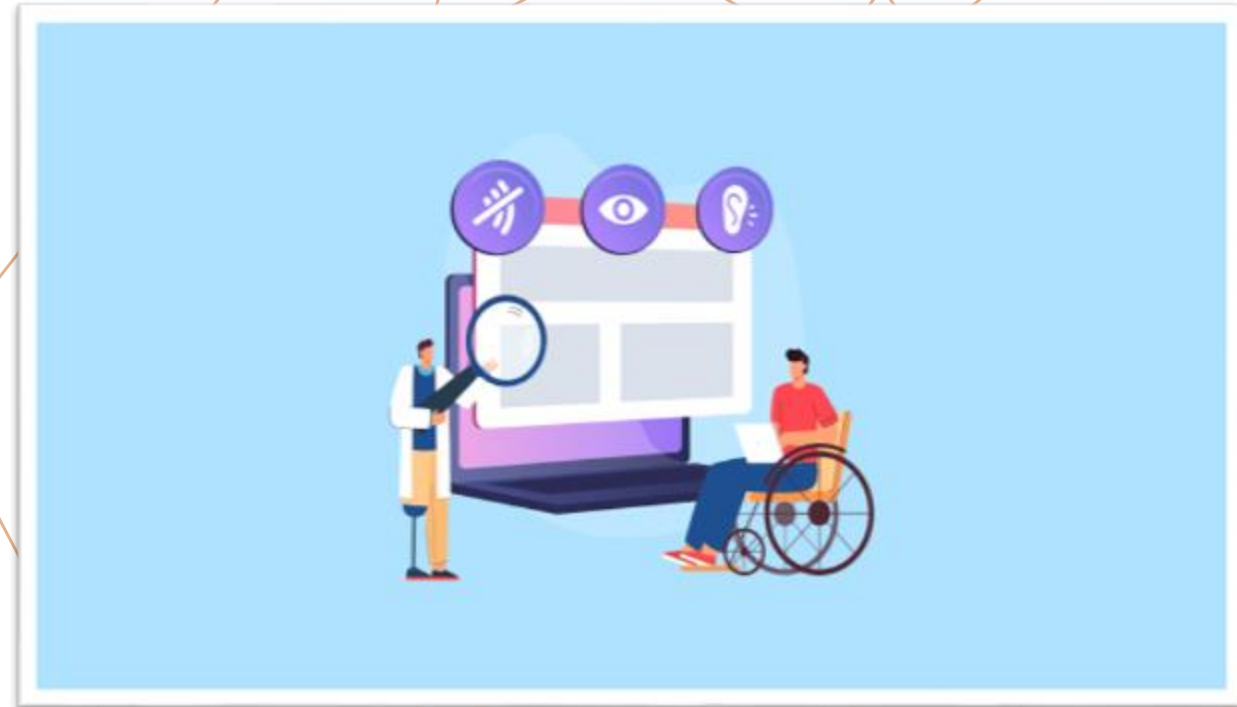


BENEFITS AND FUTURE TRENDS

- ❑ Improves software quality and user experience
- ❑ Reduces errors and usability problems
- ❑ Helps reach more users and meet legal requirements
- ❑ Future focus on early testing and better tools

CONCLUSION

- ❑ Accessibility testing is a key part of software quality assurance
- ❑ WCAG helps create inclusive and usable software
- ❑ Accessible applications benefit all users
- ❑ Accessibility should be standard practice in modern software development



A series of thin, light brown lines forming an abstract, overlapping geometric pattern on the left side of the slide. The lines intersect to create various polygonal shapes, some of which are filled with a very light brown color.

THANK YOU